GPU FinOps & Cost Optimization - Lab Report

Student: Nguyen Ngoc Tan   MSSV: 2A202600190

Notebook: gpu_finops_lab.ipynb

This report summarizes the completed GPU FinOps hands-on lab. The local Docker Compose services emulate a GPU cloud control plane, while the Kaggle notebook calls the gateway through a tunnel to monitor GPU nodes, submit workloads, record billing, evaluate spot savings, run autoscaling decisions, analyze waste, and compare real GPU training options such as FP32 and mixed precision AMP.

Area	Key result / interpretation
Cluster monitoring	The lab monitors GPU nodes, utilization, memory, power, temperature, and idle capacity.
Workload + billing	Workloads are submitted to GPUs and billing events show total cost, savings, and budget status.
Spot management	Spot capacity is cheaper than on-demand and can produce large savings, with preemption risk.
Autoscaling	Autoscaler decisions are based on utilization thresholds and cost-aware policy.
Waste analysis	Idle GPUs create waste cost; the tracker estimates waste percentage and monthly savings.
Recommendations	Main actions are using spot capacity, scheduling non-urgent jobs, and optimizing training.

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Gateway Connection Test

```
[OK] Connected to GPU FinOps Lab Gateway
Status code: 200
Response:
```


```
{
  "service": "GPU FinOps Lab Gateway",
  "endpoints": {
    "cluster": "/cluster/*",
    "billing": "/billing/*",
    "spot": "/spot/*",
    "autoscaler": "/autoscaler/*",
    "cost": "/cost/*"
  }
}
```

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Part 1 - View Cluster Nodes

```
Status: 200
Preview: {"node-00":[{"node_id":"node-00","gpu_id":0,"gpu_type":"T4","utilization":65.7664290710675,"memory_used_gb":9.083071115282362,"memory_total_gb":16.0,"power_draw_watts":45.51674000268728,"temperature_c
 Cluster has 5 nodes
============================================================

◌  node-00:
   GPU 0 [T4]  Util: 65.8% | Mem: 9.1/16.0GB | Power: 46W | Temp: 81°C
   GPU 1 [T4]  Util: 61.5% | Mem: 13.6/16.0GB | Power: 44W | Temp: 66°C

◌  node-01:
   GPU 0 [A100]  Util: 78.4% | Mem: 40.7/80.0GB | Power: 195W | Temp: 69°C
   GPU 1 [A100]  Util: 91.5% | Mem: 61.4/80.0GB | Power: 228W | Temp: 58°C

◌  node-02:
   GPU 0 [V100]  Util: 78.7% | Mem: 27.3/32.0GB | Power: 226W | Temp: 57°C
   GPU 1 [V100]  Util: 80.1% | Mem: 19.4/32.0GB | Power: 237W | Temp: 79°C

◌  node-03:
   GPU 0 [T4]  Util: 68.8% | Mem: 12.1/16.0GB | Power: 44W | Temp: 77°C
   GPU 1 [T4]  Util: 63.6% | Mem: 12.2/16.0GB | Power: 44W | Temp: 59°C

◌  node-04:
   GPU 0 [T4]  Util: 78.0% | Mem: 13.4/16.0GB | Power: 66W | Temp: 81°C
   GPU 1 [T4]  Util: 68.7% | Mem: 9.2/16.0GB | Power: 43W | Temp: 66°C
```

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Part 1 - Cluster Metrics Summary

```
Status: 200
Preview: {"total_gpus":10,"busy_gpus":10,"idle_gpus":0,"avg_utilization":73.49552444904207,"total_memory_
used_gb":218.31,"total_memory_capacity_gb":320.0,"total_power_draw_watts":1172.4,"node_count":5}
 Cluster Metrics
========================================
Total GPUs:          10
Busy GPUs:           10
Idle GPUs:           0
Avg Utilization:     73.5%
Memory Used:         218.3 GB
Memory Capacity:     320.0 GB
Total Power Draw:    1172 W
Node Count:          5
```

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Part 2 - Workload Submission

```
Submitting workloads...
  train-resnet-001: queued -> queued
  train-bert-002: queued -> queued
  inference-api-003: queued -> queued
  train-llm-004: queued -> queued

Updated metrics:
  Busy GPUs: 10/10 | Utilization: 73.5%
```

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Part 2 - Billing Summary

```
Recording billing events...
  train-resnet-001 [ON-DEMAND]: $0.0292 (saved $0.0000)
  train-bert-002 [ON-DEMAND]: $0.6117 (saved $0.0000)
  inference-api-003 [SPOT]: $0.0035 (saved $0.0082)
  train-llm-004 [SPOT]: $0.5505 (saved $1.2845)

Billing Summary:
  Total Cost:     $2.5589
  Total Savings:  $2.7078
  Budget Used:    2.6%
  Alert Status:   OK
```

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Part 3 - Spot Pricing

```
Status: 200
Preview: {"T4":{"on_demand_price":0.35,"current_spot_price":0.2644,"discount_pct":24.5,"availability":"me
dium"},"A100":{"on_demand_price":3.67,"current_spot_price":2.58,"discount_pct":29.7,"availability":"medi
 Current Spot Pricing
============================================================
GPU Type   On-Demand    Spot Price   Discount   Availability
------------------------------------------------------------
T4         $0.35       $0.2644     24.5    % medium
A100       $3.67       $2.5800     29.7    % medium
V100       $2.48       $1.7843     28.1    % medium
```

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Part 3 - Spot Request

```
Requesting Spot Instances...
  [OK] spot-t4-001 (T4): granted
  [OK] spot-t4-002 (T4): granted
  [OK] spot-a100-001 (A100): granted
```

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Part 3 - Spot Preemption and Savings

```
⚡ Simulating spot preemption event...

   Preempted instances: 1
   Still active: 5

   [!]  Preempted:
      - opt-spot-2 (ran for 4431s, 60s warning)

 Spot Savings Report:
   Spot cost:        $0.2590
   On-demand equiv:  $0.8634
   Total saved:      $0.6043 (70.0%)
```

Part 4 - Autoscaler Policy

```
Current Autoscaling Policy:
  scale_up_threshold: 70.0
  scale_down_threshold: 25.0
  cooldown_seconds: 30
  max_nodes: 10
  min_nodes: 2
  preferred_gpu_type: T4
  cost_aware: True

Updating policy...
  [OK] Policy updated
```

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Part 4 - Autoscaler Evaluation

```
Evaluating autoscaling decision...

  ⬆ Action: SCALE_UP
  Reason: Utilization 73.5% > threshold 70.0%
  Current utilization: 73.5%
  Nodes: 5 -> 6

Running 5 evaluation cycles...
  Cycle 1: no_action    | Util: 61.2% | Nodes: 6->6
  Cycle 2: no_action    | Util: 61.2% | Nodes: 6->6
  Cycle 3: no_action    | Util: 61.2% | Nodes: 6->6
  Cycle 4: no_action    | Util: 61.2% | Nodes: 6->6
  Cycle 5: no_action    | Util: 61.2% | Nodes: 6->6
```

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Part 5 - Cost Snapshots

```
Taking cost snapshots...
  Snapshot 1: Total=$0.041944 | Idle=$0.001944 | Waste=4.6%
  Snapshot 2: Total=$0.041944 | Idle=$0.001944 | Waste=4.6%
  Snapshot 3: Total=$0.041944 | Idle=$0.001944 | Waste=4.6%
  Snapshot 4: Total=$0.041944 | Idle=$0.001944 | Waste=4.6%
  Snapshot 5: Total=$0.041944 | Idle=$0.001944 | Waste=4.6%
```

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Part 5 - Waste Report

```
◌  WASTE ANALYSIS REPORT
==================================================
   Average Waste:          12.1%
   Total Idle Cost:        $0.046996
   Total Cost:             $0.401944
   Potential Monthly Save:  $1218.14
   Severity:               LOW
```

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Part 5 - Recommendations

```
 COST OPTIMIZATION RECOMMENDATIONS
============================================================

1.  [MEDIUM] USE_SPOT
   Switch fault-tolerant workloads to spot instances for 60-70% savings.
   Estimated savings: 65.0%

2.  [LOW] SCHEDULING
   Schedule non-urgent training jobs during off-peak hours for lower spot prices.
   Estimated savings: 20.0%
```

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Part 5 - Full Dashboard

```
 GPU FinOps DASHBOARD
============================================================

◌  CLUSTER: 12 GPUs across 6 nodes
   Utilization: 61.2% | Busy: 10 | Idle: 2

 BILLING: $2.5589 / $100.00 budget
   Alert: OK | Savings: $2.7078

 SPOT: Saved $0.6294 (70.0%)

◌  WASTE: 12.1% | Severity: LOW
```

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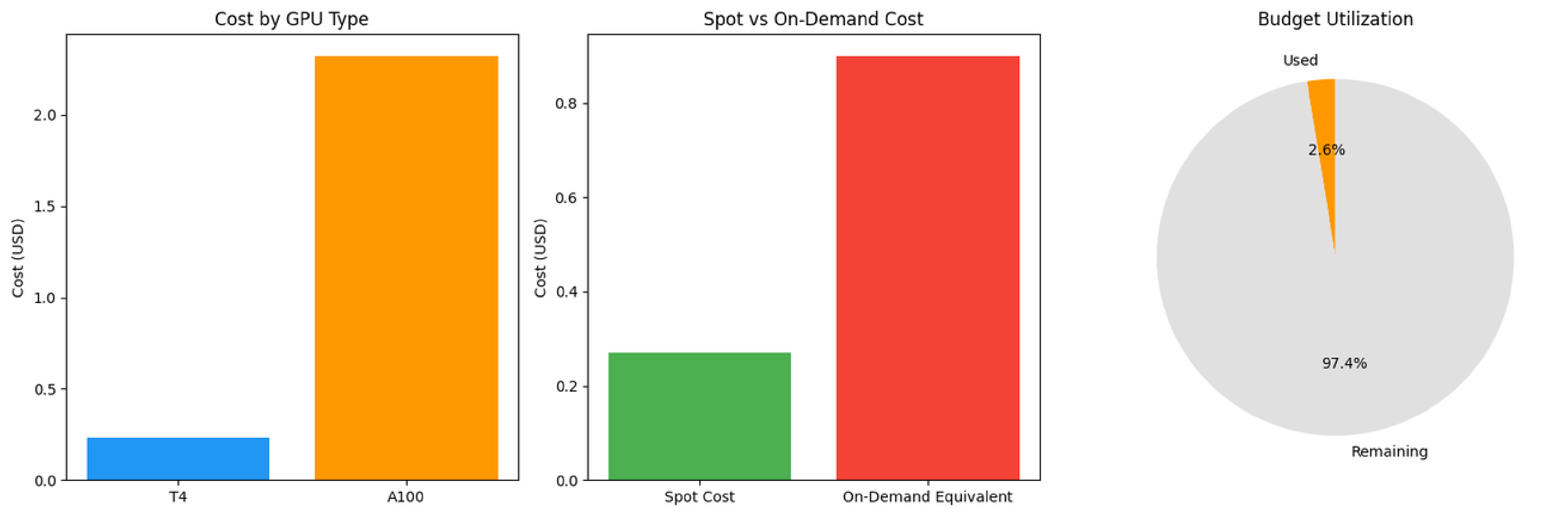
Part 6 - Cost Breakdown Visualization

```
Chart saved as finops_cost_breakdown.png
```



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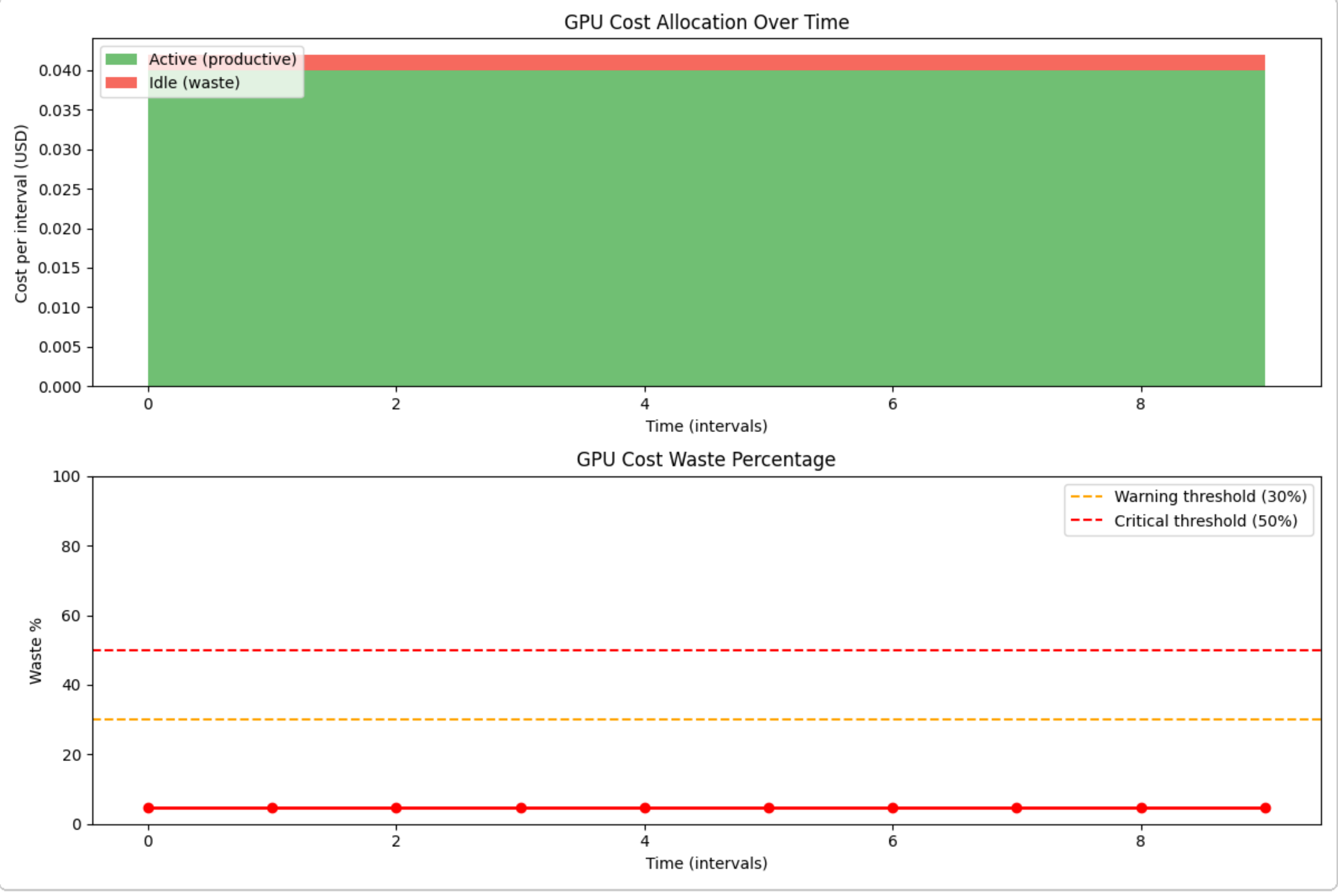
Part 6 - Time-series Cost Tracking

```
Collecting time-series data (10 snapshots)...
```



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Part 7 - Full FinOps Optimization Workflow

```
 FULL FINOPS OPTIMIZATION WORKFLOW
============================================================

1️⃣  Initial cluster state:
   GPUs: 12 | Util: 61.2% | Idle: 2

2️⃣  Submitting heavy workloads...
   After load: Util: 74.8% | Busy: 12/12

3️⃣  Autoscaler evaluation:
   Decision: scale_up - Utilization 74.8% > threshold 70.0%

4️⃣  Cost analysis:
   Total cost/interval: $0.043889
   Waste: 4.4%

5️⃣  Recommendations:
   [MEDIUM] USE_SPOT: savings ~65.0%
   [LOW] SCHEDULING: savings ~20.0%

6️⃣  Applying optimization: Switch to spot instances...
   Spot savings: $0.0451 (70.0%)

7️⃣  Final billing:
   Total spend: $2.7280
   Total saved: $2.8302
   Budget: 2.7% used

[OK] Workflow complete!
```

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Part 8 - Real GPU Detection

```
Real GPU Detected
   Name:    Tesla T4
   Memory:  15.6 GB
   Type:    T4
   Pricing: $0.35/hr (on-demand)
   CUDA:    12.8
   pynvml:  available
```

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Part 8 - GPU Metrics Diagnostic

```
==================================================
GPU METRICS DIAGNOSTIC
==================================================

1. pynvml available: True
   pynvml works! GPU util=0%, mem=472/16106 MB
   Power: 10.7W
   Temp: 42C

2. get_gpu_metrics() test:
   gpu_util_pct: 0.0
   mem_total_mb: 16106.12736
   mem_used_mb: 472.055808
   mem_util_pct: 0.0
   power_watts: 10.737
   temp_c: 42.0

   Method: pynvml

Ready for training.
```

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Part 8 - FP32 Training Summary

```
============================================================
EXPERIMENT 1: FP32 Training (Baseline)
============================================================
   Epoch 1/3 | Loss: 2.0593 | Acc: 26.8% | Time: 40.2s | Samples: 40
   Epoch 2/3 | Loss: 1.4131 | Acc: 48.1% | Time: 40.9s | Samples: 40
   Epoch 3/3 | Loss: 1.1214 | Acc: 59.8% | Time: 45.9s | Samples: 40

   Total samples collected: 120

   FP32 Summary:
   Total time:      127.1s
   Peak memory:     0.82 GB
   Avg GPU util:    96.0%
   Avg power:       67.5W
   Avg temperature: 68.5C
   Max GPU util:    99.0%
   Estimated cost:  $0.012358
```

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Part 8 - AMP Training Summary

```
============================================================
EXPERIMENT 2: Mixed Precision (AMP) Training
============================================================
   Epoch 1/3 | Loss: 1.8733 | Acc: 32.3% | Time: 20.0s | Samples: 40
   Epoch 2/3 | Loss: 1.3722 | Acc: 49.7% | Time: 19.4s | Samples: 40
   Epoch 3/3 | Loss: 1.0850 | Acc: 61.1% | Time: 19.7s | Samples: 40

   Total samples collected: 120

   AMP Summary:
   Total time:       59.1s
   Peak memory:      0.60 GB
   Avg GPU util:     91.4%
   Avg power:        65.5W
   Avg temperature: 76.2C
   Max GPU util:     94.0%
   Estimated cost:  $0.005747
```

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Part 8 - FP32 vs AMP Comparison

```
============================================================
FINOPS COMPARISON: FP32 vs Mixed Precision (AMP)
============================================================

Metric                    FP32            AMP             Improvement
----------------------------------------------------------------------
Total Time                127.1           59.1            2.15x faster
Peak Memory (GB)          0.82            0.60            0.22 GB saved
Cost (USD)                $0.012358       $0.005747       $0.006611 saved
Cost Saving %             ---             ---             53.5%
Avg GPU Util %            96.0            91.4
Avg Power (W)             67.5            65.5

--- Extrapolated Savings at Scale ---
   1 day training: FP32=$8.40 vs AMP=$3.91 -> SAVE $4.49
   1 week training: FP32=$58.80 vs AMP=$27.34 -> SAVE $31.46
   1 month training: FP32=$252.00 vs AMP=$117.19 -> SAVE $134.81
```

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Part 8 - Real GPU Cost Report

```
============================================================
REPORTING REAL GPU COSTS TO FINOPS GATEWAY
============================================================

GET https://cfeac3468cad39.lhr.life/
Status: 200
Content-Type: application/json
Response preview: {"service":"GPU FinOps Lab Gateway","endpoints":{"cluster":"/cluster/*","billing":"/bil
ling/*","spot":"/spot/*","autoscaler":"/autoscaler/*","cost":"/cost/*"}}

Gateway health: {'service': 'GPU FinOps Lab Gateway', 'endpoints': {'cluster': '/cluster/*', 'billing':
'/billing/*', 'spot': '/spot/*', 'autoscaler': '/autoscaler/*', 'cost': '/cost/*'}}

POST https://cfeac3468cad39.lhr.life/billing/record
Status: 200
Content-Type: application/json
Response preview: {"workload_id":"real-gpu-resnet18-
fp32","gpu_type":"T4","gpu_count":1,"duration_seconds":127.10727620124817,"is_spot":false,"rate_per_hour"
:0.35,"total_cost_usd":0.0124,"savings_usd":0,"project":"real-gpu-lab","timestamp":1778650391.4616735}

   FP32 workload reported:
   Cost: $0.012400 | Rate: $0.3500/hr

POST https://cfeac3468cad39.lhr.life/billing/record
Status: 200
Content-Type: application/json
Response preview: {"workload_id":"real-gpu-resnet18-
amp","gpu_type":"T4","gpu_count":1,"duration_seconds":59.10841774940491,"is_spot":true,"rate_per_hour":0.
105,"total_cost_usd":0.0017,"savings_usd":0.004,"project":"real-gpu-lab","timestamp":1778650391.952357}

   AMP workload reported (as spot):
   Cost: $0.001700 | Saved: $0.004000

POST https://cfeac3468cad39.lhr.life/cluster/workloads/submit
Status: 200
Content-Type: application/json
Response preview: {"status":"running","assigned":[{"node_id":"node-05","gpu_id":0}],"workload_id":"real-
gpu-resnet18-fp32"}

POST https://cfeac3468cad39.lhr.life/cluster/workloads/submit
Status: 200
Content-Type: application/json
Response preview: {"status":"running","assigned":[{"node_id":"node-05","gpu_id":1}],"workload_id":"real-
gpu-resnet18-amp"}

--- Updated FinOps Billing (incl. real GPU) ---

GET https://cfeac3468cad39.lhr.life/billing/summary
Status: 200
Content-Type: application/json
Response preview: {"project":"real-gpu-lab","total_cost_usd":0.0141,"total_savings_usd":0.004,"budget_usd
":100.0,"budget_remaining_usd":99.9859,"budget_utilization_pct":0.01,"total_workloads":2,"cost_by_gpu_typ
e":{"T4":{"cost":0.0141,"hours":0.05172658165295919,"workloads":2}},"alert":"OK"}
   Project:         real-gpu-lab
   Total Cost:      $0.014100
   Total Savings:   $0.004000
   Workloads:       2

POST https://cfeac3468cad39.lhr.life/cost/snapshot
Status: 200
Content-Type: application/json
Response preview: {"timestamp":1778650394.1873095,"node_costs":{"node-
00":{"total_cost":0.001944,"idle_cost":0,"gpu_count":2},"node-
01":{"total_cost":0.020389,"idle_cost":0,"gpu_count":2},"node-
02":{"total_cost":0.013778,"idle_cost":0.006889,"gpu_count":2},"node-
03":{"total_cost":0.001944,"idle_cost":0,"gpu_count":2},"node-
04":{"total_cost":0.001944,"idle_cost":0,"gpu_count":2},"node-
05":{"total_cost":0.001944,"idle_cost":0,"gpu_count":2},"node-
06":{"total_cost":0.001944,"idle_cost":0.001944,"gpu_count":2}},"tota

   Cost snapshot taken: waste=20.1%

--- FINAL DASHBOARD (Mock + Real GPU) ---

GET https://cfeac3468cad39.lhr.life/cost/dashboard
Status: 200
Content-Type: application/json
Response preview: {"cluster_metrics":{"total_gpus":14,"busy_gpus":11,"idle_gpus":3,"avg_utilization":59.5
9311393068521,"total_memory_used_gb":223.54,"total_memory_capacity_gb":384.0,"total_power_draw_watts":112
9.46,"node_count":7},"billing_summary":{"project":"default","total_cost_usd":2.728,"total_savings_usd":2.
8302,"budget_usd":100.0,"budget_remaining_usd":97.272,"budget_utilization_pct":2.73,"total_workloads":18,
"cost_by_gpu_type":{"T4":{"cost":0.40360000000000007,"hours":1.9000000000000004,"workloads":14},"A
   Total Platform Cost:  $2.7280
   Total Savings:        $2.8302
   Budget Utilization:   2.7%
   Alert:                OK
```

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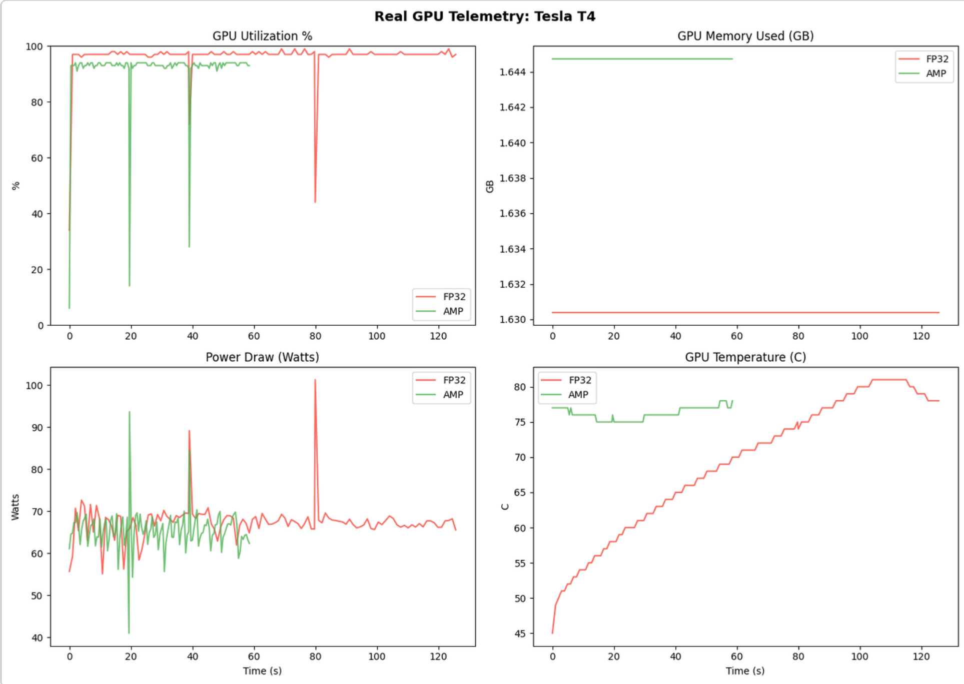
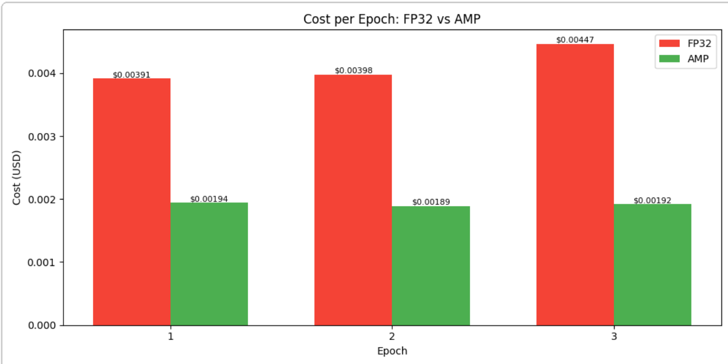
Part 8 - Real GPU Visualizations

```
Real GPU Telemetry During Training
============================================================


Charts saved: cost_per_epoch.png, real_gpu_telemetry.png
```





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Part 8.5 - Multi-GPU Cost Analysis

```
============================================================
EXERCISE 8.5.1: Multi-GPU Cost Analysis
============================================================

   gpu_count gpu_type  speedup  efficiency_pct  time_hours  total_cost_usd  \
0          1     A100     1.00           100.0    2.000000        7.340000   
1          2     A100     1.84            92.0    1.086957        7.978261   
2          4     A100     3.36            84.0    0.595238        8.738095   
3          8     A100     6.08            76.0    0.328947        9.657895   

   cost_per_speedup  
0          7.340000  
1          4.336011  
2          2.600624  
3          1.588470  

[OK] Best cost option: 1 GPU(s)
   Time: 2.00h | Cost: $7.34 | Efficiency: 100.0%
```

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Part 8.5 - Project Cost Forecast

```
============================================================
EXERCISE 8.5.2: Project Cost Forecasting
============================================================

                  phase gpu_type  gpu_count  duration_hours  base_cost_usd  \
0      Data Preparation       T4          1              40           14.0
1        Model Training     A100          4             120         1761.6
2  Hyperparameter Tuning     A100          8              60         1761.6
3      Model Evaluation       T4          2              20           14.0

   low_estimate_usd  high_estimate_usd  uncertainty_pct
0             11.90              16.10             15.0
1           1321.20            2202.00             25.0
2           1233.12            2290.08             30.0
3             12.60              15.40             10.0

[OK] Base cost:        $3,551.20
[OK] Contingency:      $710.24
[OK] Expected total:   $4,261.44
[OK] 95% confidence:   $3,573.51 - $4,949.37
```

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Part 8.5 - Optimization Opportunity Analysis

```
============================================================
EXERCISE 8.5.3: Advanced Optimization Opportunity Analysis
============================================================
Baseline cost: $1,468.00

                                name  savings_pct  estimated_savings_usd  \
0    Switch to Mixed Precision (AMP)         25.0                  367.0   
1               Optimize Batch Size         15.0                  220.2   
2           Implement Early Stopping         20.0                  293.6   
3                 Use Spot Instances         60.0                  880.8   
4  Switch to More Efficient GPU Type         40.0                  587.2   

  implementation_effort risk_level  priority_score dependencies  
0                   LOW        LOW      367.000000         None  
1                   LOW        LOW      220.200000         None  
2                MEDIUM        LOW      146.800000         None  
3                MEDIUM       HIGH      146.800000         None  
4                  HIGH     MEDIUM       97.866667         None  

Roadmap:

   step                           strategy  incremental_savings_usd  \
0     1    Switch to Mixed Precision (AMP)                 367.0000   
1     2               Optimize Batch Size                 165.1500   
2     3           Implement Early Stopping                 187.1700   
3     4                 Use Spot Instances                 449.2080   
4     5  Switch to More Efficient GPU Type                 119.7888   

   remaining_cost_usd  
0           1101.0000  
1            935.8500  
2            748.6800  
3            299.4720  
4            179.6832  

[OK] Optimized cost: $179.68
[OK] Total savings:  $1,288.32 (87.8%)
```

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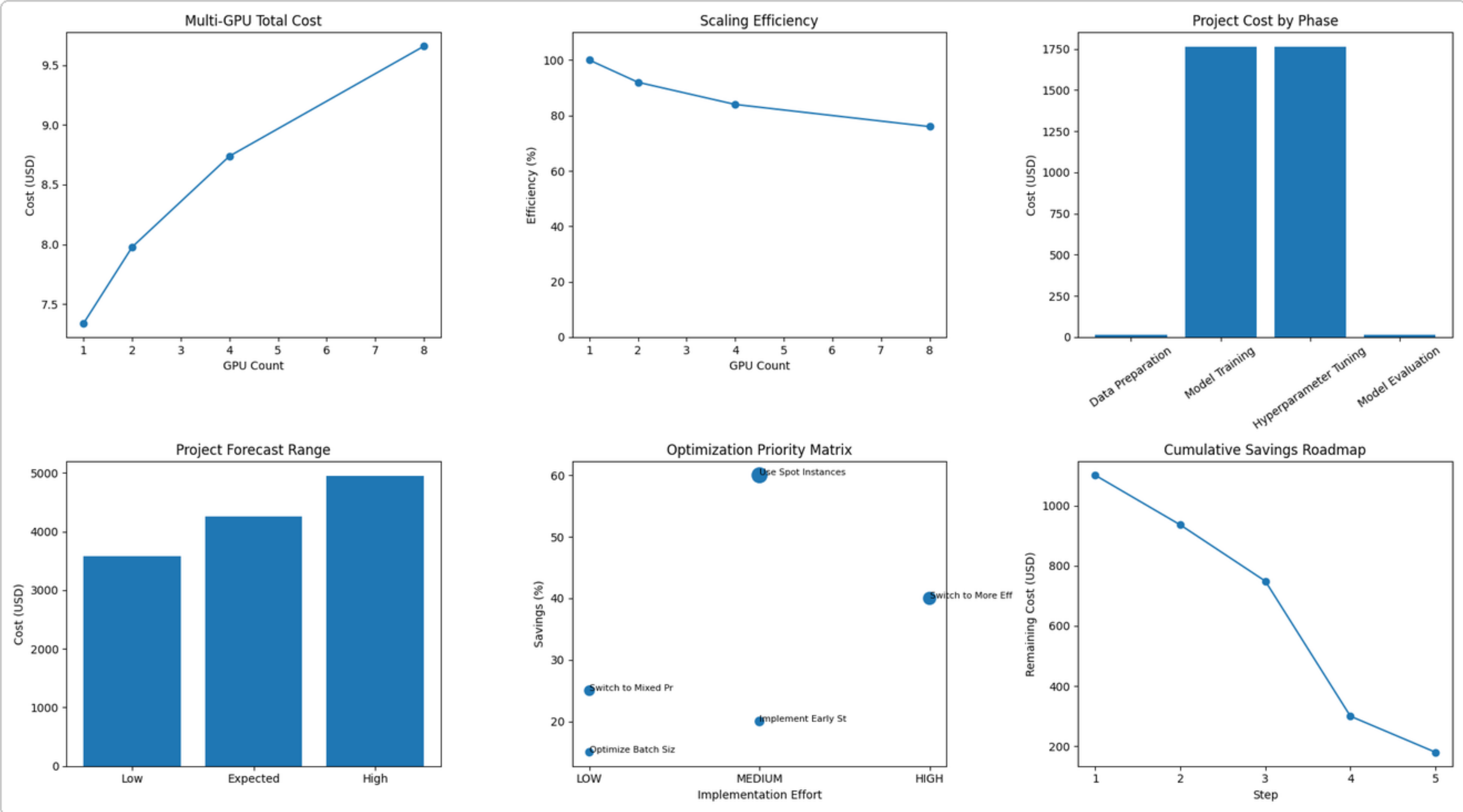
Part 8.5 - Integrated Cost Dashboard

```
============================================================
EXERCISE 8.5.4: Integrated Cost Dashboard
============================================================

[OK] Saved dashboard as advanced_finops_dashboard.png
```



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Part 8.5 - Challenge Strategy

```
============================================================
SOLUTION: Cost Optimization Strategy
============================================================

1. BASELINE COST
   GPU: 8x A100
   Duration: 200 hours
   Rate: $3.67/GPU-hour
   Baseline cost: $5,872.00

2. MULTI-GPU COST ANALYSIS
    4 GPUs | eff=0.93 | time=432.4h | cost=$6,348.11 | deadline=MISS
    6 GPUs | eff=0.88 | time=304.8h | cost=$6,710.86 | deadline=OK
    8 GPUs | eff=0.82 | time=242.4h | cost=$7,117.58 | deadline=OK
   10 GPUs | eff=0.78 | time=206.5h | cost=$7,576.77 | deadline=OK
   12 GPUs | eff=0.72 | time=183.9h | cost=$8,099.31 | deadline=OK
   16 GPUs | eff=0.62 | time=160.0h | cost=$9,395.20 | deadline=OK

   Recommended GPU count: 6 GPUs ($6,710.86, 304.8h)

3. SELECTED OPTIMIZATION STRATEGIES
   - Mixed precision / AMP: 35.0% savings, risk=LOW
     Reason: Reduces training time and memory use while preserving target accuracy.
   - Spot instances with checkpointing: 55.0% savings, risk=MEDIUM
     Reason: Uses cheaper capacity while checkpointing protects against preemption.
   - Deadline-aware GPU sizing: 0.0% savings, risk=LOW
     Reason: Chooses the cheapest GPU count that still meets the 2-week deadline.
   - Idle shutdown and job queue scheduling: 10.0% savings, risk=LOW
     Reason: Reduces idle GPU waste between experiments.
   - Frequent checkpointing and resume: 5.0% savings, risk=LOW
     Reason: Reduces lost work during spot preemption.

4. CUMULATIVE SAVINGS
   Baseline cost:             $5,872.00
   Deadline-aware plan cost:  $6,710.86
   Optimized cost:            $1,678.30
   Savings vs baseline:       $4,193.70 (71.4%)

5. COST FORECAST WITH UNCERTAINTY
   Expected cost: $1,678.30
   Low estimate:  $1,342.64
   High estimate: $2,013.96
   Confidence interval: ±20%

6. CONSTRAINT CHECK
   Under budget:       True ($1,678.30 <= $5,000.00)
   Meets deadline:     True (304.8h <= 336h)
   Preemption risk OK: True
   Accuracy target OK: True

7. STRATEGY SUMMARY
   Decision: APPROVED
   Recommended plan:
   - Use 6x A100
   - Switch from FP32 to AMP/mixed precision
   - Use spot instances with checkpointing
   - Enable idle shutdown and queue-based scheduling
   - Track cost daily through FinOps dashboard

   Justification:
   The original plan costs $5,872.00, which is above the $5,000.00 budget. The deadline-aware GPU plan
   uses 6x A100 and finishes in 304.8h, within the 336h deadline. After AMP, spot, idle shutdown, and
   checkpointing, the expected cost is $1,678.30, while keeping preemption risk within the MEDIUM limit
   and preserving the minimum accuracy target of 0.95.
```

Conclusion

The lab demonstrates the core GPU FinOps loop: gain visibility into cluster utilization, associate workloads with cost, choose cheaper capacity such as spot where safe, use autoscaling to control idle capacity, and use waste analysis to prioritize optimization work. The real GPU training section also shows that code-level optimizations such as mixed precision can reduce runtime and cost, complementing infrastructure-level optimizations.

Deliverables generated in this package: report.pdf, screenshots/, generated_charts/, and notebook/gpu_finops_lab.ipynb.